

Link Budget: $P_r[dB] = P_t[dB] + \underbrace{G_t[dB] + G_r[dB] - (L_p[dB] + \chi[dB])}_{\text{large scale gain } G}$

Path Loss: $L_P(D) = K_{ref} \left(\frac{D}{D_{ref}} \right)^\eta \rightarrow L_P(D)[dB] = K_{ref}[dB] + 10\eta \log_{10} \left(\frac{D}{D_{ref}} \right)$

Free Space Path Loss: $L_P(D)[dB] = 20 \log_{10} \left(\frac{4\pi D}{\lambda_c} \right)$

Fading: $f_\chi(\xi) = \frac{1}{\sqrt{2\pi}\sigma\xi} \exp \left\{ -\frac{(\log_e \xi)^2}{2\sigma^2} \right\} \xrightarrow{dB} f_\chi(\xi) = \frac{1}{\sqrt{2\pi}\sigma_{dB}} \exp \left\{ -\frac{\xi^2}{2\sigma_{dB}^2} \right\}$

Rice: $f_{|h|}(\xi) = 2(K+1)\xi e^{-(K+1)\xi^2-K} I_0 \left(2\sqrt{K(K+1)}\xi \right)$

Rayleigh Magnitude: $f_{|h|}(\xi) = \xi e^{-\frac{1}{2}\xi^2}$

Exponential magnitude squared: $f_{|h|^2}(\xi) = e^{-\xi}$

Doppler: $\nu = \underbrace{\frac{v}{c} f_c}_{\nu_M} \cos(\theta)$

Coherence: $N_c = \frac{T_c}{T} = BT_c \xrightarrow{\text{IID Blocks}} \frac{1}{2\nu_m T}$

$T_c \gg T_s \quad T_c = \frac{1}{2\nu_m} \propto \frac{1}{\nu_m}, \quad \nu_m \ll B$

PDP: $\mu_\tau = \int \tau S_h(\tau) d\tau \rightarrow \sum_{q=0}^{Q-1} \tau_q P_q$

$\sigma_\tau^2 = T_d = \sqrt{\int (\tau - \mu_\tau)^2 s_h(\tau) d\tau} \rightarrow \sqrt{\sum_{q=0}^{Q-1} (\tau_q - \mu_q)^2 P_q}$

Flat

$$\boxed{\text{Frequency Flat iff: } \tau_m \ll T_s \iff T_d \ll T_s \iff B_c \gg B}$$

$$\implies \tau_i(t) \approx \tau_j(t) \approx \hat{\tau} \forall i, j \rightarrow h(\tau) = h\delta(\tau)$$

LMSSE Estimator: $W^{MMSE} = \frac{1}{\sqrt{G}} \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$

$E = \frac{E_s}{N_t} I - \frac{E_s}{N_t} D^* T^* \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$

Some Z

$$H(z) = \frac{1}{1 - az^{-1}} \rightarrow h[n] = (a)^n u[n] \quad \text{ROC: } |z| > |a|$$

$$w[n] = x[n - n_0] \xrightarrow{Z} W(z) = z^{-n_0} X(z) \quad R_W = R_X$$

LMMSE: $W_{MMSE} = R_y^{-1} R_{ys}$

$E = R_s - R_{ys}^* R_y^{-1} R_{ys}$

Gauss Vec MMSE: $\hat{s}(y) = \sqrt{\rho} R_s A^* (\rho A R_s A^* + R_z)^{-1} y$

$E = (R_s^{-1} + \rho A^* R_z^{-1} A)^{-1}$